Assignment 3

(Questions from Module 5)

Name 1	
Matric No.	
Section	
Date	

Name 2	
Matric No.	
Section	
Date	

Total Marks: 40

Instructions

- Students must answer **ALL questions** with complete workings, including conversions, arithmetic, and floating-point operations.
- Hardware algorithms should show full iteration tables with proper binary notation and indicate overflow, normalisation, truncation, or rounding.
- Complete this assignment in **pairs** (**2 members ONLY**) and **submit handwritten** work in **PDF format** through eLearning within 7 days, as specified by your instructor.

Collaboration Requirements

Work together with your assigned partner and ensure both students contribute equally to the assignment. Communicate clearly and divide tasks fairly from the start.

Submission Format

Write neatly by hand in black or blue ink on standard A4 paper, then scan and convert to PDF. Include your names, student ID numbers, course details, instructor name, and submission date on the cover page.

Deadline and Submission

Confirm your specific 7-day deadline with your instructor. Submit only through the eLearning platform and keep a personal copy for your records.

QUESTION 1 (15 marks)

a. Consider a 16-bits system uses odd parity. Show the calculation for 9280h + 903Ah in binary and determine the value for each of the flags in Table 1. [6m]

Table 1

ZF	SF	CF	OF	AF	PF

b. ISA (Instruction Set Architecture) format was defined as the following format:

4 *bit*	12 *bit*
Opcode	Operand

Opcode: 0001_2 = MOV instruction, 1000_2 = SUB instruction

Trace the execution of instructions for address 40Ah shown in Table 2. Show all the changes in CPU registers (control and general purpose registers) and clearly write the *micro-operations* with related to the instructions in Table 3. [9m]

Table 2

Memory Address	Memory Content (hex)	Instructions
409	1003	`MOV BX,3h`
40A	8743	`ADD COA,BX`
40B		
…	…	…
743	0005	`COA`
744	0F00	

Table 3

Clock	IP/PC	MAR	MBR	IR	BX	COA	Micro-Operations
t_0	409				0003	0005	IP/PC = 409

QUESTION 2 (15 marks)

a. There are 6 segments as follows:

Segment Name	Segment Execution Time (ns)
Fetch Instruction (FI)	57
Decode Instruction (DI)	25
Calculate Operand (CO)	45
Fetch Operand (FO)	35
Execute Instruction (EI)	45
Write Operand (WO)	40

The interface delay is 5 ns.

i. What is the execution time for 1500 instructions without pipeline? [2m]
ii. What is the execution time for 1500 instructions with pipeline? [2m]
iii. What is the maximum/theoretical speedup? [1m]
iv. What is the real speedup? [2m]

b. A pipeline processor has 5 segments - Fetch Instruction (FI), Decode Instruction (DI), Fetch Operand (FO), Execute Instruction (EI) and Store Operand (SO). There are five instructions (I1 – I5) needed to be executed:

I1: CMP EAX, ECX
I2: ADD TOTAL, 2
I3: JMP I5
I4: ADD EAX, 10H
I5: SUB ECX, 20H

Table 4: The space time diagram for question i and ii (template)

	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11

i. Fill in the space time diagram for instructions I1 – I5. Show which instructions need to be flushed (cross the boxes involved). [4m]
ii. Using Delay Branch (NOP) to solve the problem of Branching, show in new space time diagram the solution. [4m]